

Internet Programming 622

Name and Surname: _____

Student ITS No: _____

Qualification: _____ Year of Study: _____ Semester: _____

Assignment due date: _____ Date submitted: _____

QUESTION	EXAMINER MARKS	MODERATOR MARKS	REMARKS

ASSIGNMENT INSTRUCTIONS

Please tick each box to confirm completion.

Use Times New Roman font, size 12, with 1.5 line spacing throughout the document. Apply Harvard Referencing Style for all citations and references.

For essay-style assignments, please include the following sections:

Table of Contents Introduction

Main Body (with relevant subheadings) Conclusion

References

Submit the assignment in PDF format on Moodle. Use the specified cover page provided.

Include a signed declaration of originality.

DECLARATION OF ORIGINALITY:

I hereby declare that this assignment is my own work and has not been copied from any other source except where due acknowledgment is made. I affirm that all sources used have been properly cited and that this submission complies with the institution's policies on academic integrity and plagiarism.

Student Signature: _____

Date: _____

QUESTIONS ONE

[60 MARKS]

The explosive growth of social media over the past decade has radically changed how individuals and businesses communicate, share content, and establish their presence in the digital world. Whether it's Facebook connecting friends and family, Instagram promoting brands through photos, TikTok reshaping video content, or X (formerly Twitter) transforming public discourse, these platforms have one thing in common — they enable real-time, interactive communication with global reach. In this assignment, you are tasked with building a fully functional social media platform, from scratch, using PHP and MySQL as your backend technologies, and HTML5, CSS3, JavaScript, or jQuery for the front-end. You are prohibited from using any design frameworks such as Bootstrap, Tailwind CSS, Materialize, or content management systems like WordPress. This project will test not only your technical skills but also your ability to design and implement a secure, interactive, and user-friendly online platform.

Your social platform must allow users to register by entering their full name, email, password, and an optional profile picture. The email address must be unique in the system to prevent duplicate accounts. All user data must be stored securely in a MySQL database, and the password must be hashed using PHP's `password_hash()` function before being stored. Users who already have an account should be able to log in using their email and password. Upon successful login (after verifying credentials from the database), users should be redirected to a personalized dashboard. The login page should also provide a "Forgot Password" option where users can reset their password using their registered email. Session handling using `$_SESSION` must be used to manage login states and protect restricted pages from unauthorized access.

Once inside the system, each user should be able to access a timeline/dashboard that shows their recent activity. Users must be able to post text updates and optionally upload an image with each post. These posts should be stored in the database and displayed in reverse chronological order. Each post must also include a timestamp. In addition to the timeline, you must implement a user profile page that displays the user's profile picture, full name, and a summary of all their posts. Users must have the ability to edit or update their profile picture and display name. Furthermore, users must be able to search for other users in the system

using a search bar. When a user visits another user's profile, they should be able to view that user's public timeline, including posts and profile picture, but not have the ability to edit them.

An essential part of your platform is a private messaging system that allows users to communicate with one another. This feature should enable users to search for another user by name or email, initiate a conversation, and send direct text messages. All messages must be stored in a separate table within the database and displayed in conversation format. Users should only be able to see messages sent to or received by them. This feature will demonstrate your understanding of one-to-one relationships in MySQL and user access control in PHP.

To enhance the user experience, use JavaScript or jQuery to add interactivity, such as form validation, live search suggestions, image previews before upload, or message notifications. You are encouraged to demonstrate creative front-end techniques, including the use of responsive layouts, custom CSS styles, and basic animations or effects. All elements must be built from scratch — do not copy code or use prebuilt UI kits. Security is a critical part of the assessment: your system must include measures against SQL injection, form tampering, and XSS. Any dynamic data must be properly escaped, and all queries must be parameterized using prepared statements (PDO or mysqli).

Finally, the project must be neatly organized into folders (e.g., `includes/`, `css/`, `js/`, `uploads/`) and well-commented to demonstrate clean and maintainable code. You must submit a .zip file containing all source code, your database .sql file, and a README.txt or userguide.pdf file explaining how to run the system, create a test account, and test the main features. You may also include optional features like notifications, friend requests, or account settings for bonus marks, but all core features must function properly to pass.

Expected outputs: Use your own details (name, Picture, posts, email address, etc)



Cleve

Email: cleve.davey@gmail.com

Cleve's Posts



Cleve

pp



2024-09-29 21:22:22



Cleve

II



Search Users:

Search by name or email

Search



Cleve

cleve.davey@gmail.com

Send a Message

Username of the recipient

Type your message...

Send

Your Messages:

Cleve

To: Joe Rogan

Hi Joe

2024-09-29 12:21:02





Joe Rogan

Share Something ...



POST

Login

Wrong username/password combination

Username

Password

Login

Not yet a member? [Sign up](#)

Register

Password is required
Username already exists
Email already exists

Username

Cleve

Email

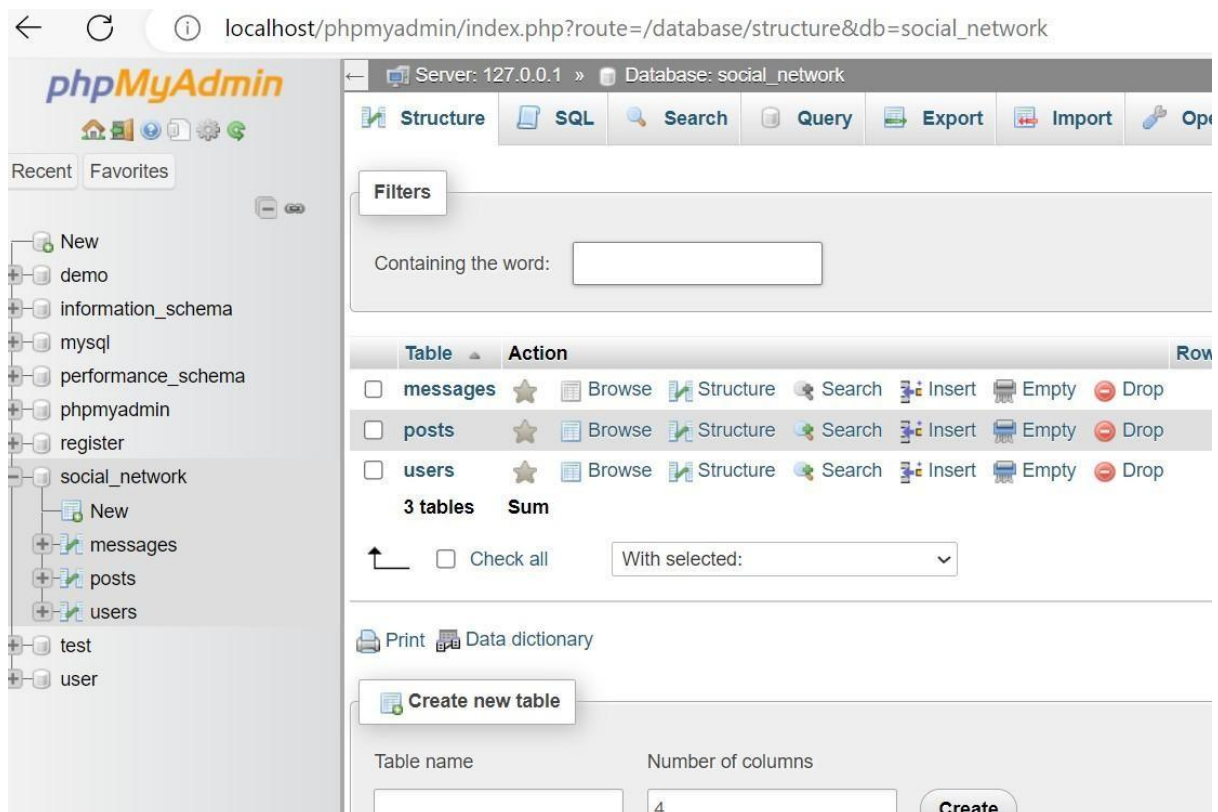
cleve.davey@gmail.com

Password

Confirm password

Register

Already a member? [Sign in](#)



QUESTIONS TWO

[30 MARKS]

A warehousing company based in Cape Town, South Africa, is the latest victim of 'CEO fraud' which is basically a type of cybercrime in which the attacker impersonates a CEO or other company executive using a fake email address. Assuming that the company consulted you as IT consultant for a solution to prevent this crime from ever occurring in their organization:

2.1 Discuss in detail, various CEO fraud attack methods, and various ways of preventing this type of cybercrime. [10]

2.2 Develop an IT Security policy for Sam Dunn Enterprises [10]

2.3 How does PHP handle data processing and form submissions, and what are some of the key considerations and challenges involved in implementing secure and scalable form processing in a real-world web application? (10)